

Helix Thready — System Architecture (Area Index)

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Rev	Date	Author	Change
1	2026-07-21	swarm (System Architecture)	Initial area index — files, deps, gap coverage, open items
2	2026-07-21	swarm (review pass)	reading-order <code>.mmd</code> sibling + multi-para explanation; split GAP 2.9 rows; PROC-4
3	2026-07-22	swarm (Pass 3 depth)	Register new <code>post-lifecycle.md</code> capstone (homes <code>post-lifecycle.mmd</code>); update reading order + diagram-sources; note consistency fixes (concurrency <code>retrying</code> -status, security sensitive-content multi-para)
4	2026-07-22	swarm (Pass 3 depth, cont.)	Close CAT-1/CAT-2 (source-verified <code>rag</code> / <code>cache</code> / <code>messaging</code> <code>pkg</code> / layouts; corrected <code>SkillRegistry = dev.helix.agent/skillregistry & MCP= digital.vasic.mcp</code> import paths); rewrite \$7 register with accurate CLOSED/OPEN status; refresh \$8 verified inventory; record the diagram-depth pass (system-overview C4 ×2, data-flow ×2, processing-pipeline, asset-download, component-deps, event-flow split to multi-paragraph)
5	2026-07-22	swarm (critic consistency)	Register <code>post-lifecycle.md</code> in \$3 docs table + \$4 reading-order (embedded diagram was out of sync with its <code>.mmd</code> sibling; prose said “eleven documents”); complete \$5 diagram-sources table (add <code>sticky-invalidation</code> , <code>retry-state-machine</code> , <code>post-lifecycle</code> — three existing <code>.mmd</code> files were unlisted); cross-refs to the service-discovery diagram fix (registry→broadcast) and PROC-2 closure

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1. Purpose

This is the canonical entry point for the **System Architecture** area of Helix Thready. It delivers the enterprise-grade, implementation-ready architecture: system overview (C4 context+container), the full component catalog with in-house submodule mapping, data-flow and event model, concurrency/idempotency, service discovery, security, the four core subsystems (messenger ingestion, processing pipeline, semantic search, assets & download), and the `post-lifecycle.md` capstone that walks one post end-to-end across every plane. Every document follows `../CONVENTIONS.md` exactly and never contradicts the authoritative sources (the final answered request, the gap register, the original request).

2. Upstream / downstream dependencies

Upstream (this area consumes):

- **Authoritative sources of truth** (read-only, private): the final answered request (`helix_thready_research_request_final.md` — decision matrix §0.2, architecture §2, workflow §3, Q1–Q45 §18), the subsystem gap register (`helix_thready_subsystem_gaps_and_improvements.md`), and the original request. These are never contradicted.
- **In-house engines** (`vasic-digital` / `HelixDevelopment` submodules) whose real interfaces anchor the code examples — see §8.

Downstream (these areas derive from this one):

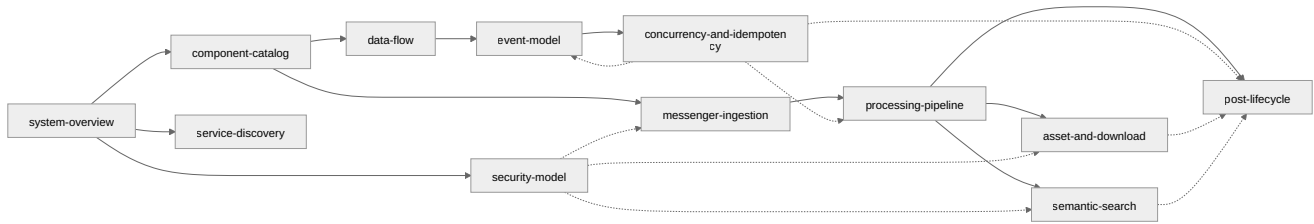
- **api/** — the REST `/v1` OpenAPI 3.1, WebSocket/SSE event contract, and SDK strategy all realize the component boundaries, event catalog and endpoints defined here.

- **database/** — the full ERD, PostgreSQL + pgvector DDL, indexes, partitioning and migrations expand the architecture-level data model in [data-flow.md](#) and the DDL fragments in [concurrency-and-idempotency.md](#) and [semantic-search.md](#).
- **deployment/** — rootless Podman Compose, 3 envs, discovery/ports, Let's Encrypt, backup/DR consume [service-discovery.md](#) and the scale drivers in [system-overview.md](#).
- **testing/** — the 15 mandated test types target the contracts, events and TDD skeletons seeded throughout this area.

3. Documents in this area

Document	Scope
system-overview.md	C4 context + container; scale/SLO/tenancy drivers; cross-cutting concerns
component-catalog.md	Every service & reused submodule; in-house mapping; maturity; dependency matrix
data-flow.md	End-to-end flow; architecture ERD; SoR vs derived; partitioning; sensitive flow
event-model.md	EventBus/JetStream; one-time vs sticky + invalidation; at-least-once; durable replay; event catalog
concurrency-and-idempotency.md	Single-claim per post; retry/back-off; circuit breaker; precedence
service-discovery.md	discovery + mdns + port_prefix; dynamic ports; health; subdomain routing
security-model.md	auth; three-tier RBAC; encryption at rest/in transit; sensitive-content handling
messenger-ingestion.md	Herald + gotd/td; Max adapter; ThreadReader root+organic-reply assembly
processing-pipeline.md	Hashtag classification; Skill dispatch order; content-type recipe matrix; OCR
semantic-search.md	Lumen-style: embeddings → pgvector → rag; HelixLLM /v1/embeddings; HashEmbedder guard
asset-and-download.md	Asset Service (Catalogizer); Download Manager; Boba/MeTube; standardized callback
post-lifecycle.md	Capstone: end-to-end ingest→claim→process→finalize sequence; stage/event/failure reference; latency-SLO decomposition

4. Reading order



Rendered PNG/SVG exported via Docs Chain (§11.4.65). Source: `diagrams/reading-order.mmd`.

Explanation (for readers/models that cannot see the diagram). The diagram is a directed reading graph over the twelve documents of this area. Start with the system overview for the C4 mental model, then the component catalog to learn which engines back which services. From the catalog, follow the data plane along the solid spine: data-flow → event-model → concurrency-and-idempotency — how a post moves, how events behave, and how exactly-once is enforced. These three are ordered because each consumes the previous one’s vocabulary (the event model names the events the data-flow produced; the concurrency model dedupes the deliveries the event model admits).

Two cross-cutting documents branch directly off the overview rather than off the data spine: service-discovery (how services find each other and get deterministic ports) and security-model (auth, three-tier RBAC, encryption, sensitive-content handling). They are drawn as separate branches because they apply to every service uniformly, not to a single point in the flow. The functional spine then runs messenger-ingestion → processing-pipeline, which fans out to semantic-search and asset-and-download — the two subsystems the pipeline delegates to. The pipeline also flows into the **post-lifecycle** capstone, which is the terminus of the graph: it stitches ingestion, claim, processing, assets and search into one end-to-end sequence, so it is drawn as a sink that the concurrency, asset and semantic docs also feed (the dashed edges into PL).

The dashed edges encode the cross-cutting dependencies that are not part of the linear reading order: security underpins ingestion, assets and search (session secrets, sealed content, tenant scoping), and the concurrency model underpins both the pipeline and the event model (single-claim makes at-least-once safe). The post-lifecycle capstone is best read **last** — it assumes the vocabulary every other document establishes and re-derives none of it. Any document can be read standalone — each carries full context and its own gap/open-item registers — but following the arrows minimizes forward references.

5. Diagram sources

All Mermaid diagrams are embedded in the docs **and** saved as sibling `.mmd` sources in `diagrams/` so Docs Chain can export PNG/SVG later `[CONSTITUTION §11.4.65]`:

.mmd	Used in
diagrams/reading-order.mmd	index (this file)
diagrams/c4-context.mmd	system-overview
diagrams/c4-container.mmd	system-overview
diagrams/component-deps.mmd	component-catalog
diagrams/data-flow-e2e.mmd	data-flow
diagrams/data-model.mmd	data-flow
diagrams/event-flow.mmd	event-model
diagrams/sticky-invalidation.mmd	event-model (§4.1)
diagrams/single-claim.mmd	concurrency-and-idempotency
diagrams/retry-state-machine.mmd	concurrency-and-idempotency (§5.1)
diagrams/discovery.mmd	service-discovery
diagrams/rbac.mmd	security-model
diagrams/sensitive-content.mmd	security-model
diagrams/thread-assembly.mmd	messenger-ingestion
diagrams/processing-pipeline.mmd	processing-pipeline
diagrams/semantic-search.mmd	semantic-search
diagrams/asset-download.mmd	asset-and-download
diagrams/post-lifecycle.mmd	post-lifecycle

6. Gap-register coverage summary

Every gap-register item relevant to System Architecture is addressed with a design plan or a tracked `[BUILD-NEW]` item, tagged `[GAP: ...]` in the owning document. None claims a stub “works”.

Gap	Priority	Headline	Addressed in
2.1	P0	HelixLLM default embedder is non-semantic HashEmbedder	semantic-search.md
2.5	P2	LLMsVerifier port :7061 vs :8080	semantic-search.md, component-catalog.md
2.6	P0	VisionEngine has no OCR engine	processing-pipeline.md
2.7	P1	Embeddings has no native llama.cpp backend	semantic-search.md
2.8	P2	Memory search is word-overlap, not semantic	semantic-search.md
2.9 (claim)	P1	session_orchestrator claim registry is design-only	concurrency-and-idempotency.md
2.9 (token-opt)	P1	token_optimizer WIP / TOON scaffold — token-optimization mandate [§11.4.198]	processing-pipeline.md §5.1
3.1	P1	VectorDB only pgvector wired; others unverified	semantic-search.md
3.2	P1	database has no partitioning/sharding helpers	data-flow.md
4.1	P0	helix_skills has no processing/execution engine	processing-pipeline.md
5.1	P0	Herald MTPROTO in QA harness; Max empty stub	messenger-ingestion.md
6.1	P1	Catalogizer not decoupled; Streaming = WS hub	asset-and-download.md
6.2	P0	filesystem: no HTTP source, no download semantics	asset-and-download.md
6.3	P0	Download Manager does not exist	asset-and-download.md
6.4	P1	Boba callback contract bespoke	asset-and-download.md
6.5	P0	MeTube poll-only, no outbound webhook	asset-and-download.md
6.6	P1		

Gap	Priority	Headline	Addressed in
		No standardized callback/ task module	asset-and-download.md, event- model.md
7.1	P2	Searchable-but-sealed credentials	security-model.md
7.2	P1	auth JWT default HMAC- SHA256 (needs asymmetric)	security-model.md
7.3	P0 (mobile)	Security-KMP mobile storage is in-memory stub	security-model.md
7.4	P1	Auth-KMP must consume fixed Security-KMP	security-model.md
11.*	P0–P1	New subsystems (Asset/ Download/User/Max/OCR/ webhook/callback/EventBus svc/ThreadReader/ Semantic)	component-catalog.md §4 + owning docs

7. Consolidated open-items register

Each `[OPEN: ...]` is a tracked workable item, not a papered-over gap. Most were source-verification of a FLAGGED interface (gap register §13 re-verification backlog) before implementation — and the Pass-3 source reads have now **closed** the interface-verification items (their owning docs carry the matching `[CLOSED: ...]`). What remains open is genuinely *deferred decisions* (deployment topology, tuning, a build-time schema), never an unverified “it works”. The Status column is authoritative; each item’s full detail lives in its owning doc’s Open-items section.

ID	Item	Status	Owner doc
OVERVIEW-1/2	HTTP/3 gateway topology; JetStream env isolation	OVERVIEW-1 OPEN (deployment); OVERVIEW-2 resolved — env lives in <code>SubjectPrefix</code> (event-model §3)	system-overview / event-model
CAT-1/2	rag/messaging/cache layouts; MCP/SkillRegistry import paths	CLOSED — layouts source-verified; paths corrected to <code>digital.vasic.mcp</code> / <code>dev.helix.agent/skillregistry</code>	component-catalog §8
EVT-1/2	JetStream retention window; sticky compacted-stream vs KV	EVT-1 narrowed — <code>LimitsPolicy</code> + <code>FileStorage</code> verified, only window sizing is a deployment knob; EVT-2 OPEN (impl choice)	event-model
CONC-1/2	<code>models.BackgroundTask</code> dedup field; breaker package path	CONC-1 CLOSED — no <code>DedupeKey</code> / <code>ErrDuplicateTask</code> , dedup is Thready-side; CONC-2 narrowed — <code>resilience.Manager</code> verified, unify-vs-split is a design call	concurrency-and-idempotency
DISC-1/2	<code>discovery.Registry</code> API; reverse-proxy choice	DISC-1 CLOSED — no <code>Registry</code> ; real API is <code>pkg/broadcast (Announcer / Listener / Responder)</code> ; DISC-2 OPEN (proxy)	service-discovery
SEC-1/2/3	<code>security/pkg/policy</code> API; redaction spec; internal mTLS vs tokens	SEC-1 CLOSED — <code>Enforcer</code> / <code>EvaluateAll</code> (most-restrictive); SEC-2/3 OPEN (spec/decision)	security-model
ING-1/2/3	Max OneMe port; herald model mapping; forum-topic normalization	ING-2 CLOSED — <code>commons_messaging/channels (Channel / IsSelfEcho)</code> ; ING-1/3 OPEN (research/mapping)	messenger-ingestion
PROC-1/2/3/4	Canonical <code>SKILL.md</code> schema; SkillRegistry/MCP paths; per-Skill caps;	PROC-2 resolved by CAT-2 (paths source-confirmed); PROC-1/3/4 OPEN (schema/tuning/scope)	processing-pipeline

ID	Item	Status	Owner doc
	token_optimizer/ TOON scope		
SEM-1/2/3	vectordb/embeddings interfaces; local code- embedding GGUF; HNSW tuning	SEM-1 CLOSED — <code>VectorStore</code> / <code>EmbeddingProvider</code> verified; SEM-2/3 OPEN (GGUF/tuning)	semantic-search
ASSET-1/2/3	Asset Service extraction plan; transcoder choice; callback auth	OPEN (design/decision — owned by the BUILD-NEW submodules)	asset-and-download
DF-1/2/3	ERD ownership (database area); partition granularity; chunk partitioning	OPEN (owned by database/ testing packs)	data-flow
PL-1/2/3	Per-stage soft-timeout budgets summing to 30 min; end-to-end integration harness; Max full-lifecycle	PL-1/2 OPEN (load-test/testing pack); PL-3 NOTE — Max lifecycle blocked on <code>[GAP: 5.1.2]</code> OneMe port	post-lifecycle

8. Verified-at-source module inventory

To honor the “in-house first, do not invent APIs” rule, these interfaces were **read at source** this pass (via `gh api`) and anchor the code examples:

- `vasic-digital/EventBus` — `pkg/event` (`Event` , `Handler` , `Subscription`), `pkg/bus` (`Config` , `New`), `pkg/nats` (`JetStream` `Bus` with `Publish` / `Subscribe` / `Close` , `ensureStream` , default stream `EVENTBUS`). **VERIFIED.**
- `vasic-digital/BackgroundTasks` — `interfaces.go` (`TaskQueue` , `TaskExecutor` , `ProgressReporter` , `TaskRepository` with `Dequeue(workerID, maxCPU, maxMem)` , `StuckDetector`), `models/task_status.go` (status enum). **VERIFIED.**
- `vasic-digital/herald` — `pkg/messenger/messenger.go` (`Messenger` , `Factory` interfaces); `pkg/messenger/max.go` (empty stub, `// TODO: MAX platform` — confirms `[GAP: 5.1.2]`). **VERIFIED.**
- `vasic-digital/discovery` — `pkg/report` (`Report` , `scanner.Service`). **VERIFIED.**
- `vasic-digital/port_prefix` — `portprefix.Exposed(prefix, internalPort, taken)` , band mapping. **VERIFIED.**
- `vasic-digital/security` — `pkg/securestorage/securestorage.go` (`Storage` interface, `FileStorage` AES-256-GCM, credential/token/key helpers); `pkg/policy.Enforcer` (`EvaluateAll` most-restrictive `Deny>Audit>Allow` , `Policy` / `Rule` / `Condition`). **VERIFIED** (`[CLOSED: SEC-1]`).

- `vasic-digital/RAG` / `cache` / `Messaging` — `pkg/` layouts read (`rag` : chunker/retriever/ reranker/hybrid/pipeline; `cache` : memory/redis/postgres/distributed/policy/ service; `messaging` : broker/producer/consumer/kafka/rabbitmq). **VERIFIED** (Pass 3 cont., `[CLOSED: CAT-1]`).
- `vasic-digital/MCP_Module` = module `digital.vasic.mcp` , `pkg/` {server,client,protocol,registry,adapter,config} ; `vasic-digital/SkillRegistry` = module `dev.helix.agent/skillregistry` (flat package: `executor.go` / `loader.go` / `manager.go` / `registry.go` / `storage.go` / `validator.go` / `types.go`). **VERIFIED** (`go.mod` read, `[CLOSED: CAT-2]`) — the previously-quoted `digital.vasic.mcp_module` / `digital.vasic.skill_registry` paths were **wrong** and are corrected.
- `vasic-digital/VectorDB` (`pkg/client.VectorStore` + `Vector` / `SearchQuery` / `SearchResult` / `DistanceMetric` ; backends pgvector/qdrant/pinecone/milvus) and `vasic-digital/Embeddings` (`pkg/provider.EmbeddingProvider` ; providers openai/voyage/jina/google/cohere/bedrock — no `llama` / `hash` , confirming `[GAP: 2.7]`). **VERIFIED** (`[CLOSED: SEM-1]`).
- `vasic-digital/herald` current `commons_messaging/channels.Channel` (`SendReplyGeneric` / `BotSelfIdentity` / `DownloadAttachment`) + self-filter `IsSelfEcho` / `StampSender` (`SelfIdentity` / `IdentityKind`). **VERIFIED** (`[CLOSED: ING-2]`).
- `vasic-digital/discovery` `pkg/broadcast` (`Announcer` / `Listener` / `Responder` / `ServiceInfo`) + `pkg/resilience.Manager` (`Connected/Disconnected/Reconnecting/Offline`). **VERIFIED** (`[CLOSED: DISC-1]` , `CONC-2` narrowed).

Every interface the Wave-1 pass had left unread — `vectordb` `VectorStore` , `embeddings` provider, `security/pkg/policy` , herald's current channel model, discovery's real API, and the `rag` / `cache` / `messaging` /MCP/SkillRegistry layouts — has since been **read at source** across the Pass-3 depth passes (each has the matching `[CLOSED: ...]` in its owning doc). What remains representative rather than source-exact is confined to the `[BUILD-NEW]` composition code (ThreadReader, Skill Dispatch Engine, Download Manager, standardized callback module, thin Event Bus Service, Semantic-search service), which is Thready-new orchestration by definition and carries its own `[GAP: ...]` / `[OPEN: ...]` tags — no reused engine's interface is now shown as a guess.

9. Conventions & provenance

Provenance tags used throughout `[CONVENTIONS §3]` : `[CONSTITUTION §x]` authoritative · `[IN-HOUSE: module]` reuse existing submodule · `[RESEARCH]` web/source research · `[OPERATOR]` operator decision · `[DEFAULT — adjustable]` proposed default · `[BUILD-NEW]` confirmed new-submodule gap · `[GAP: id]` addresses a gap-register item. Every diagram carries a multi-paragraph prose explanation and a sibling `.mmd` ; every file carries the metadata header, revision table and ToC.

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